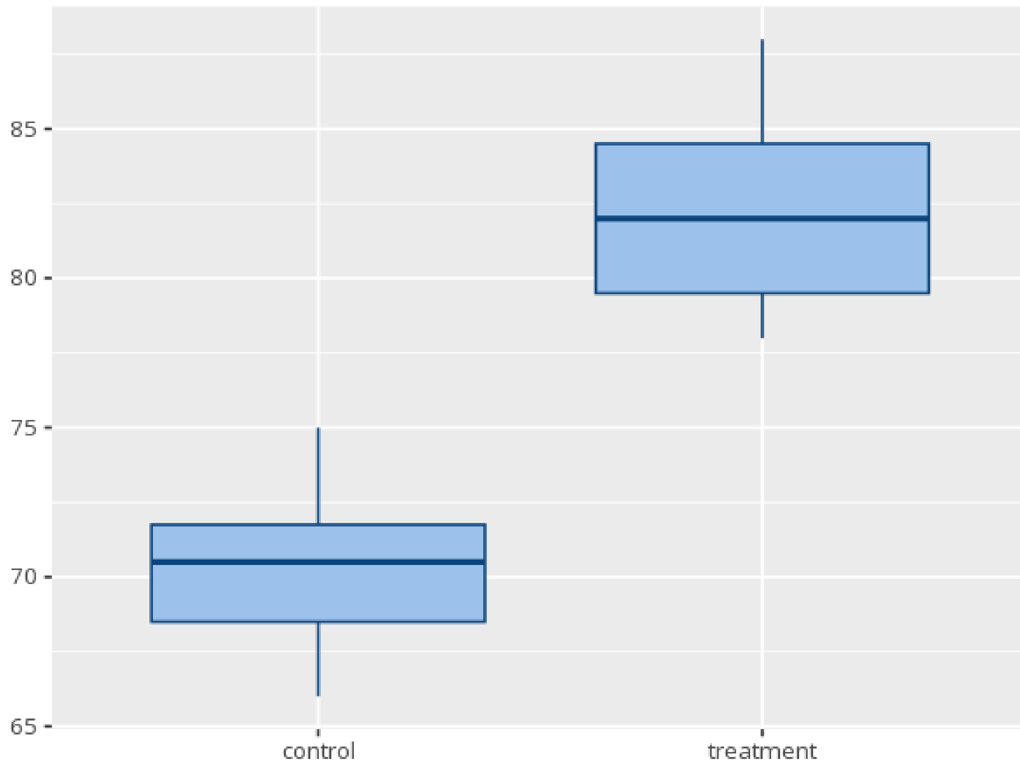
1 Mann-Whitney U

Group	N	Mean rank	Median	IQR	Sum of ranks
control	6	3.50	70.50	3.25	21.00
treatment	6	9.50	82.00	5.00	57.00

U	Z	p	r [95% CI]
0	−2.88	.002	−1.00 [−1.00, −
Hodges-Lehmann median difference = −12.00, 95% CI [−17.00, −7.00]			



The nonparametric counterpart to the independent t-test, comparing whole distributions via ranks rather than means. A p below alpha means one group's values tend to be systematically higher or lower (stochastic dominance) — this equals a difference in medians only when the two groups have similar distribution shapes. r gives the effect size. Your significance threshold (α) is 0.05. Method: Wilcoxon rank sum exact test.

A Mann-Whitney U test gave U=0, Z=−2.88, p = .002, r=−1.00 [−1.00, −1.00].

Statistical basis: Mann-Whitney U test (Mann, H. B., Whitney, D. R. (1947). "On a test of whether one of two random variables is stochastically larger than the other." Annals of Mathematical Statistics, 18(1), 50-60.); Rankbiserial effect size r (Ben-Shachar MS, Lüdecke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.).